

Saturated Distributed Centroid Tracking Control of Cluster Formation and Its Application to VTOL AAVs

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Abstract—To improve flexibility and adaptability of the multi-agent formation, this paper studies the centroid tracking problem of cluster formation. In particular, a team of agents are separated into multiple clusters, and the clustered agents are coordinated such that i) each cluster forms a prescribed configuration, ii) all the cluster centroids form another prescribed configuration, and iii) the centroid of the whole team tracks a reference trajectory. The agents are characterized by second-order dynamics. In addition, heterogenous input saturation levels are considered into the multi-agent system. A saturated distributed control algorithm resorting to three distributed observers is proposed. The first two distributed observers with proper initial conditions enable each agent to accurately obtain the centroid of its corresponding cluster and that of the whole team. And the third distributed observer helps each agent get the accurate reference acceleration information. It is demonstrated that the proposed saturated distributed control algorithm guarantees the accomplishment of the centroid tracking objective of cluster formation. Besides, we apply the proposed saturated distributed control algorithm to the clustered VTOL AAV formation. Simulation results finally verify the effectiveness of the proposed saturated distributed control algorithm.

Index Terms—Distributed control, distributed observer, formation, multi-agent system, saturation.

I. INTRODUCTION

ALONG with the rapid development of informatics, communication, sensor and other related disciplines, multi-agent formation has already become a promising research topic among academic and engineering communities, which has been extensively applied in military and civil areas [1], [2], [3], [4]. In a typical formation maneuver, all the agents are coordinated in consistent with prescribed constraints on their states. Benefiting from its potential advantages of flexibility, scalability, and robustness, the multi-agent formation has been studied substantially [5]. Among numerous existing schemes,

the distributed control, which effectually avoids unnecessary over-centralization, turns out to be the prominent one.

Instead of the rigidity-based formation that depends on strict (infinitesimally) rigid conditions to ensure the global uniqueness of the configuration [6], [7], the position-displacement-based formation without too many constraints are more popular. Particularly, the internal agents are driven to their corresponding target positions of a pre-specified configuration uniquely characterized by position displacement constraints. Hitherto, a number of distributed control algorithms have been proposed for the position-displacement-based formation of general multi-agent systems or further actual vehicle teams. For instance, by designing a prescribed-time observer to estimate the states of the leader, a distributed control algorithm was proposed in [8] for the accurate formation tracking of second-order multi-agent systems within a user-determined prescribed time. Another distributed adaptive formation control algorithm was proposed in [9] for the leader-follower multi-agent system subject to measurement noises under dynamically changing topologies. Besides, a distributed control algorithm introducing a prescribed performance strategy was designed in [10] for the leader-follower multi-agent formation with prescribed performance guarantees. Moreover, for unmanned aerial/ground vehicles (*abbr.* autonomous aerial vehicles (AAVs)/UGVs) [11], [12], [13], [14], [15], [16], [17] and unmanned surface vessels (*abbr.* USVs) [18], [19], [20], multiple feasible distributed control algorithms were proposed for their leader-follower formation tracking. However, the basic leader-follower formation framework in [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] is relatively rigorous in the sense that the formation configuration offset with respect to reference is heavily dependent on the leader. In practice, to circumvent this rigorous restriction, it is desirable to enable the formation centroid instead of the leader to track the reference. However, how to complete the formation centroid tracking is tricky especially when the centroid information is unavailable for each agent that has just access to the information of its neighbors. Although a distributed control algorithm was proposed in [21] to solve the formation centroid tracking problem of a cluster of AAVs, the control framework therein is relatively simple with less flexibility.

In addition, input saturation is another noteworthy issue when it comes to practical controlled objects. If not duly tackled, it may degrade the control performance or even destabilize the controlled system [22]. Especially for multi-agent systems, the

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performance degradation of one agent will badly affect the whole team by the manner of network diffusion. To surmount the dilemma caused by the input saturation when coordinating the multi-agent system, researchers have begun to append saturation constraints to the distributed control algorithms. For general multi-agent systems, a saturated distributed control algorithm was proposed in [23] for the global consensus by combining linear system and Lyapunov theories. To robustify the mixed effect of data quantization, aperiodic sampling, and data quantization, the multi-agent practical set consensus was achieved in [24] by leveraging the properly designed saturated distributed control algorithm. Moreover, another viable distributed algorithms under the event-triggered driven framework were developed in [25], [26], [27]. In addition, several saturated distributed control algorithms were proposed in [28], [29], [30] for the formation tracking of multiple AAVs, where the boundedness of the thrust force can be guaranteed. For more flexible and efficient formation maneuvers, another distributed control algorithms were built in [31], [32], [33] for non-holonomic vehicles with the saturated control. Nevertheless, the saturation levels considered in [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] are symmetric for individuals and identical for different individuals. Due to mechanical aging, fabrication imprecision and other unpredictable factors, such symmetry and identity can hardly be maintained. By considering heterogeneous saturation levels, a saturated sampled-data distributed control algorithm was proposed in [34] for first-order multi-agent systems. However, whether it can be directly extended to high-order systems is questionable. Furthermore, although the saturated distributed control using heterogeneous saturation levels achieves the leader-follower multi-agent tracking in [35], it is just applicable to the case where the leader moves in a constant velocity and cannot be extended to solving the formation centroid tracking problem.

Instead of the conventional leader-follower formation tracking problem that depends on the leader to decide the configuration displacement with respect to the reference trajectory, this paper studies the centroid tracking problem of cluster formation for second-order multi-agent systems. Particularly, multiple clusters comprise a whole team, and the agents therein cooperate such that each cluster forms a prescribed configuration, all the cluster centroids form another prescribed configuration, and the centroid of the whole team tracks the reference trajectory. Such a form of multi-agent formation just makes demands on the cluster centroids while the dependence on some agents is released. This greatly improves flexibility and adaptability of the multi-agent formation. In addition, to get closer to actual physical objects, each agent suffers from input saturation constraints. More practically, we consider heterogeneous saturation levels, which means that the saturation level of each agent is asymmetric and those of different agents are non-identical. A saturated distributed control algorithm leveraging three viable distributed observers is proposed. Benefiting from the first two distributed observers, each agent is capable of estimating the centroid of its belonging cluster and that of the whole team accurately. Besides, the third distributed observer guarantees that each agent estimates the reference acceleration accurately. It is shown that the proposed saturated distributed control algorithm enables the multi-agent system to accomplish the centroid tracking objective of

cluster formation. Furthermore, we extend the proposed saturated distributed control algorithm to clustered VTOL AAVs and show their accomplishment of the centroid tracking objective of cluster formation. Compared with the existing works, the main contributions are enumerated as follows.

- i) Rather than the leader-follower multi-agent formation in [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] that depends on the leader to determine the configuration displacement with respect to the reference trajectory, the centroid tracking of cluster formation studied in this paper use the cluster centroids. This effectively relieves the reliance on the leader.
- ii) Besides the formation centroid tracking task as done in [21], the configuration construction herein is related to both intra-cluster agents and inter-cluster configuration centroids. This multi-cluster separation framework is beneficial to improving the flexibility and adaptability of the large-scale multi-agent system.
- iii) Compared with the saturated distributed control algorithms in [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] that just consider symmetric and identical input saturation levels, the proposed one allows the heterogeneous saturation levels for multi-agent systems with second-order dynamics.

The remainder of this paper is arranged as follows. Section II describes the centroid tracking problem of cluster formation. Section III presents the main results including the saturated distributed control algorithm design and the detailed stability analysis. Section IV applied the proposed saturated distributed control algorithm to the clustered VTOL AAV formation. Section V verifies the centroid tracking effectiveness via simulation. And Section VI draws final conclusions.

II. PROBLEM STATEMENTS

A. Clustered Multi-Agent System

Consider a large-scale multi-agent system of $N \geq 2$ agents, each of which is modeled by the following second-order dynamics:

$$\dot{p}_j = v_j, \quad (1a)$$

$$\dot{v}_j = u_j, \quad (1b)$$

where $p_j, v_j \in \mathbb{R}^d$ denote the generalized position and velocity variables, and $u_j \in \mathbb{R}^d$ denotes the generalized input variable. All the agents are separated into $n \geq 2$ disjoint clusters, labeled by $\mathcal{M} = \{1, 2, \dots, n\}$, each of which consists of $m_i \geq 1$ agents, labeled by $\mathcal{V}_i = \{i_1, i_2, \dots, i_{m_i}\}$. Hence, $\sum_{i=1}^n |\mathcal{V}_i| = N$, where $|\cdot|$ denotes the set cardinality. The communication relationship within each cluster i is characterized by an undirected graph $\mathcal{G}_i = \{\mathcal{V}_i, \mathcal{E}_i\}$ with $\mathcal{E}_i \subseteq \mathcal{V}_i \times \mathcal{V}_i$. Particularly, $(i_j, i_k) \in \mathcal{E}_i \Leftrightarrow (i_k, i_j) \in \mathcal{E}_i$ means that the intra-cluster agents i_j and i_k can communicate with each other; and they are thus called intra-cluster neighbors. Besides, let set $\mathcal{N}_j^{intra}$ include the intra-cluster neighbors of agent j . Moreover, we authorize one agent in each cluster as a correspondent that is responsible for communicating with other clusters. Without loss of generality, i_1 serves as the correspondent for cluster i . We define set

$\bar{\mathcal{V}} = \{1_1, 2_1, \dots, n_1\}$ to include all the correspondents and set $\bar{\mathcal{E}} \subseteq \bar{\mathcal{V}} \times \bar{\mathcal{V}}$ to include all the inter-cluster communication links. Also, let set $\mathcal{N}_j^{inter}$ include the inter-cluster neighbors of agent j . Obviously, $\mathcal{N}_j^{inter} = \emptyset$ if $j \notin \bar{\mathcal{V}}$.

The overall communication relationship over the network can be described by a topology graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ with $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2 \cup \dots \cup \mathcal{V}_n$ and $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2 \cup \dots \cup \mathcal{E}_n \cup \bar{\mathcal{E}}$. Besides, let set $\mathcal{N}_j = \mathcal{N}_j^{intra} \cup \mathcal{N}_j^{inter}$ include all the neighbors of agent j . Moreover, graph $\mathcal{G}$ is assumed to be connected, which means that there exists a path from each agent to every other agent. Additionally, we introduce an adjacency matrix $D = [d_{jk}] \in \mathbb{R}^{N \times N}$ to quantify the communication relationship, i.e., $d_{jk} > 0$ (or $= 0$) if $(k, j) \in \mathcal{E}$ (or $(k, j) \notin \mathcal{E}$). Also, we define another Laplacian matrix $L = [l_{jk}] \in \mathbb{R}^{N \times N}$ as $l_{jk} = -d_{jk}$ for $j \neq k$ and $l_{jj} = \sum_{k=1, k \neq j}^N d_{jk}$.

Remark 1: Given a connected overall graph $\mathcal{G}$, it can be trivially shown that each intra-cluster graph $\mathcal{G}_i$ is also connected.

B. Centroid Tracking of Cluster Formation

For each cluster i , its configuration centroid is defined as

$$p_i^c = \frac{1}{m_i} \sum_{j=1}^{m_i} p_{ij}. \quad (2)$$

Upon this basis, the configuration centroid of the whole team is further defined as

$$p^c = \frac{1}{n} \sum_{i=1}^n p_i^c. \quad (3)$$

We focus on the distributed control algorithm design in this paper to coordinate the clustered multi-agent system (1) such that

- i) the agents in each cluster are capable of forming a pre-specified configuration;
- ii) the configuration centroids of the clusters are also capable of forming a pre-specified configuration; and
- iii) the configuration centroid of the whole team is capable of tracking a reference trajectory $p^r \in \mathbb{R}^d$.

We refer to this kind of formation tracking maneuver as *centroid tracking of cluster formation*. Mathematically, the centroid tracking objective of cluster formation is composed by the following components:

- intra-cluster configuration construction:

$$\lim_{t \rightarrow \infty} (p_j(t) - p_k(t)) = \delta_{jk}, \quad j, k \in \mathcal{V}_i, \quad i \in \mathcal{M}, \quad (4)$$

- inter-cluster configuration construction:

$$\lim_{t \rightarrow \infty} (p_i^c(t) - p_j^c(t)) = \theta_{ij}, \quad i, j \in \mathcal{M}, \quad (5)$$

- centroid tracking:

$$\lim_{t \rightarrow \infty} (p^c(t) - p^r(t)) = 0_d, \quad (6)$$

- velocity tracking:

$$\lim_{t \rightarrow \infty} (v_j(t) - \dot{p}^r(t)) = 0_d, \quad j \in \mathcal{V}, \quad (7)$$

where $\delta_{jk} \in \mathbb{R}^d$ denotes the expected displacement between intra-cluster agents j and k , and $\theta_{ij} \in \mathbb{R}^d$ denotes the expected displacement between the centroids of clusters i and j . The

expected displacements δ_{jk} and θ_{ij} should be chosen carefully such that no collision occurs in the desired configuration. Besides, it is not necessary to make all the agents know the reference trajectory information, while it is only broadcast to one or some certain agents.

In practice, for a large-scale multi-agent system, it is favorable to divide them into several clusters according to some feasible guidelines for the improvement of its flexibility and adaptability. The separation guidelines are commonly made based on the concerned mission types or agent-related roles. In such a case, the studied centroid tracking of cluster formation is of more interest than the conventional leader-follower formation tracking. Moreover, for the studied centroid tracking of cluster formation, the configuration of each cluster with respect to the reference trajectory is determined by its centroid instead of the leader. This thus effectively relieves its reliance on the leader and any agent can be the leader. In addition, the studied centroid tracking of cluster formation has a strong adaptability to the expansion or contraction of the network scale in that it allows one or more agents/clusters to leave or join the formation at any time. In spite of the above advantages of the centroid tracking of cluster formation, as far as a large-scale actual multi-agent system is concerned, there still exist two challenges for its achievement, which are elaborated as follows.

- *Centroid unavailability:* Constrained by finite communication channels and limited range of communication, it is impossible for each agent to receive information from any other agent over a large-scale network even though they are in the same cluster. This leads to a dilemma where each agent cannot obtain the configuration centroid of its corresponding cluster, and thus has no idea of the configuration centroid of the whole team. Therefore, we are confronted with a great challenge in the fulfillment of the centroid-related sub-objectives (5) and (6).
- *Input saturation constraint:* Due to the confined capabilities of physical actuators, input saturation constraint is inevitable for practical autonomous agents. The resultant nonlinearity, once neglected, will result in serious performance descent or even destabilize the system. On the other hand, subject to craft imprecision, mechanical aging or other factors, it is difficult to sustain saturation symmetry of each agent or saturation uniformity of different agents. However, the non-uniform and asymmetric saturation levels lay a challenge in the distributed control algorithm design, especially for the achievement of multiple sub-objectives (4)–(7).

In the subsequent section, a saturated distributed control algorithm that are tolerant against non-uniform and asymmetric saturation levels is designed for the clustered multi-agent system (1) such that the centroid tracking objective (4)–(7) of cluster formation is achieved.

III. MAIN RESULTS

In this section, resorting to three effective distributed observers, a saturated distributed control algorithm is first designed. Then, under the Lyapunov framework, the fulfillment

of the centroid tracking objective (4)–(7) of cluster formation is demonstrated in detail.

A. Saturated Distributed Control Algorithm Design

Since the reference trajectory is only known to a part of agents, to guarantee that the configuration centroid of the whole team tracks the reference trajectory, it is essential for the agents that know the reference trajectory information to figure out the configuration centroid. Accordingly, we first introduce a distributed observer for each agent $j \in \mathcal{V}_i$ to estimate the configuration centroid of its belonging cluster i :

$$\begin{aligned} \ddot{\psi}_j = & - \sum_{k \in \mathcal{N}_j^{intra}} d_{jk}(\hat{p}_j - \hat{p}_k - (p_j - p_k - \delta_{jk})) \\ & - \sum_{k \in \mathcal{N}_j^{intra}} d_{jk}(\hat{v}_j - \hat{v}_k - (v_j - v_k)), \end{aligned} \quad (8)$$

where two auxiliary variables ψ_j and $\dot{\psi}_j$ are initialized by

$$\sum_{j \in \mathcal{V}_i} \psi_j(0) = 0_d, \quad \sum_{j \in \mathcal{V}_i} \dot{\psi}_j(0) = 0_d, \quad (9)$$

d_{jk} is the (j, k) -entry of the adjacency matrix D , and $\hat{p}_j = p_j + \psi_j$, $\hat{v}_j = v_j + \dot{\psi}_j$ are the position and velocity estimates of the configuration centroid of cluster i .

Having obtained the configuration centroid of its corresponding cluster for each agent, we next introduce another distributed observer for it to further estimate the configuration centroid of the whole team:

$$\begin{aligned} \ddot{\eta}_j = & - \sum_{k \in \mathcal{N}_j} d_{jk}(\hat{p}_j^c - \hat{p}_k^c - (\hat{p}_j - \hat{p}_k - \bar{\theta}_{jk})) \\ & - \sum_{k \in \mathcal{N}_j} d_{jk}(\hat{v}_j^c - \hat{v}_k^c - (\hat{v}_j - \hat{v}_k)), \end{aligned} \quad (10)$$

where two auxiliary variables η_j and $\dot{\eta}_j$ are initialized by

$$\sum_{j=1}^N \eta_j(0) = 0_d, \quad \sum_{j=1}^N \dot{\eta}_j(0) = 0_d, \quad (11)$$

$\hat{p}_j^c = \hat{p}_j + \eta_j$, $\hat{v}_j^c = \hat{v}_j + \dot{\eta}_j$ are the position and velocity estimates of the configuration centroid of the whole team, and $\bar{\theta}_{jk} = \theta_{jk}$ if agents j and k are in the distinct clusters, and otherwise $\bar{\theta}_{jk} = 0_d$.

Now that the information of the configuration centroid of the whole team has been available, we can now move on with the control design. To begin with, we specify two composite variables for each agent $j \in \mathcal{V}$ as follows:

$$\begin{aligned} s_j = & \sum_{k \in \mathcal{N}_j^{intra}} d_{jk}(p_j - p_k - \delta_{jk}) + \sum_{k \in \mathcal{N}_j^{inter}} d_{jk}(\hat{p}_j - \hat{p}_k - \theta_{jk}) \\ & + b_j(\hat{p}_j^c - p^r), \end{aligned} \quad (12)$$

$$r_j = \sum_{k \in \mathcal{N}_j} d_{jk}(v_j - v_k) + b_j(v_j - \dot{p}^r), \quad (13)$$

where $b_j > 0$ if agent j knows the reference trajectory information, and $b_j = 0$ otherwise. Using these two composite variables,

we propose a saturated control for each agent $j \in \mathcal{V}$ as follows:

$$u_j = -\text{sat}_j(c(s_j + r_j)) - \text{sat}_j(cr_j) + \hat{\rho}_j, \quad (14)$$

where $\text{sat}_j(x) = [\text{sat}_j(x_1), \text{sat}_j(x_2), \dots, \text{sat}_j(x_d)]^T$ for $x = [x_1, x_2, \dots, x_d]^T$ with each saturation function $\text{sat}_j(\cdot)$ defined by

$$\text{sat}_j(\cdot) = \begin{cases} \bar{\alpha}_j, & \text{if } (\cdot) \geq \bar{\alpha}_j, \\ (\cdot), & \text{if } \bar{\alpha}_j > (\cdot) \geq \underline{\alpha}_j, \\ \underline{\alpha}_j, & \text{if } (\cdot) < \underline{\alpha}_j, \end{cases} \quad (15)$$

$\bar{\alpha}_j > 0$, $\underline{\alpha}_j < 0$ are upper and lower saturation bounds, $c > 0$ is a control parameter, $\hat{\rho}_j$ is the estimate of the reference acceleration. Herein, the saturation level of each agent is allowed to be asymmetric, i.e., $\bar{\alpha}_j \neq -\underline{\alpha}_j$, while the saturation levels of distinct agents are not necessarily uniform, i.e., $\bar{\alpha}_j \neq \bar{\alpha}_k$ or $\underline{\alpha}_j \neq \underline{\alpha}_k$. Besides, the estimate $\hat{\rho}_j$ is in the form of $\hat{\rho}_j = \lambda_j \tanh(\sigma_j)$, where σ_j is an auxiliary variable, $\tanh(\sigma_j) = [\tanh(\sigma_{j,1}), \tanh(\sigma_{j,2}), \dots, \tanh(\sigma_{j,d})]^T$, and $\lambda_j \geq \sup_{t \geq 0} \|\ddot{p}^r(t)\|$ is a constant parameter with $\|\cdot\|$ as the 2-norm of a vector. To obtain the auxiliary variable σ_j , we design a distributed observer in the following form:

$$\begin{aligned} \dot{\sigma}_j = & -\Lambda_j^{-1} \left[\frac{\kappa_j}{\lambda_j} \left(\sum_{k \in \mathcal{N}_j} d_{jk}(\hat{\rho}_j - \hat{\rho}_k) + b_j(\hat{\rho}_j - \ddot{p}^r) \right) \right. \\ & \left. + \frac{\hat{\beta}_j}{\lambda_j} \tanh \left(\frac{\sum_{k \in \mathcal{N}_j} d_{jk}(\hat{\rho}_j - \hat{\rho}_k) + b_j(\hat{\rho}_j - \ddot{p}^r)}{\varepsilon(t)} \right) \right], \end{aligned} \quad (16)$$

where $\Lambda_j = \text{diag}\{\text{sech}^2(\sigma_{j,1}), \text{sech}^2(\sigma_{j,2}), \dots, \text{sech}^2(\sigma_{j,d})\}$, $\kappa_j > 0$ is a constant parameter, $\varepsilon(t) = \exp(-\epsilon t)$ with $\epsilon > 0$, and $\hat{\beta}_j$ with $\hat{\beta}_j(0) > 0$ is an adaptive parameter updated by the following adaptive law:

$$\begin{aligned} \dot{\hat{\beta}}_j = & \left(\sum_{k \in \mathcal{N}_j} d_{jk}(\hat{\rho}_j - \hat{\rho}_k) + b_j(\hat{\rho}_j - \ddot{p}^r) \right)^T \\ & \tanh \left(\frac{\sum_{k \in \mathcal{N}_j} d_{jk}(\hat{\rho}_j - \hat{\rho}_k) + b_j(\hat{\rho}_j - \ddot{p}^r)}{\varepsilon(t)} \right). \end{aligned} \quad (17)$$

To facilitate the implementation of the proposed saturated distributed control algorithm, the signal flow diagrams of the distributed observers (8) with (9), (10) with (11), (16) with (17) and the saturated control (14) are illustrated in Figs. 1–4. Moreover, using the proposed saturated control (14), each component is upper and lower bounded by $2 \max\{\bar{\alpha}_j, -\underline{\alpha}_j\} + \lambda_j$ and $2 \min\{-\bar{\alpha}_j, \underline{\alpha}_j\} - \lambda_j$, respectively. In practice, we can choose appropriate bounds $\bar{\alpha}_j$ and $\underline{\alpha}_j$ for each agent such that its available control does not violate a particular input saturation constraint anytime.

Remark 2: To overcome the dilemma caused by the unavailability of the centroid information and reference acceleration information, three distributed observers (8), (10), and (16) are designed. To implement these three distributed observers as well as the proposed saturated control (14), each agent $j \in \mathcal{V}$ needs

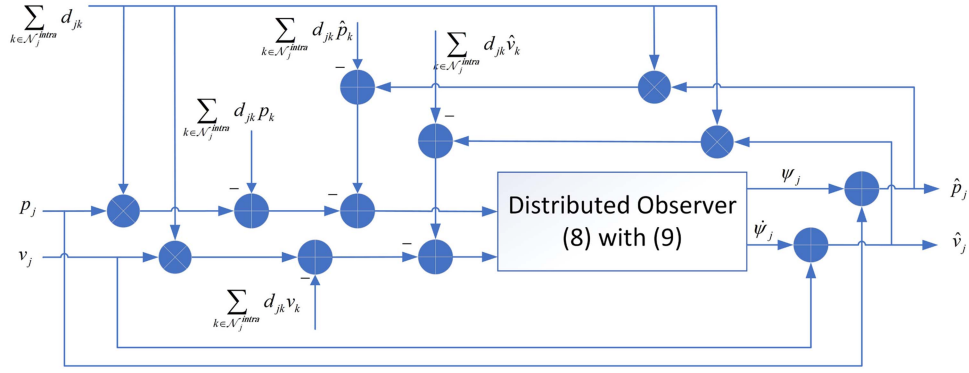


Fig. 1. Signal flow diagram of distributed observer (8) with (9) (Inputs: p_j , v_j and p_k , v_k , $k \in \mathcal{N}_j^{intra}$).

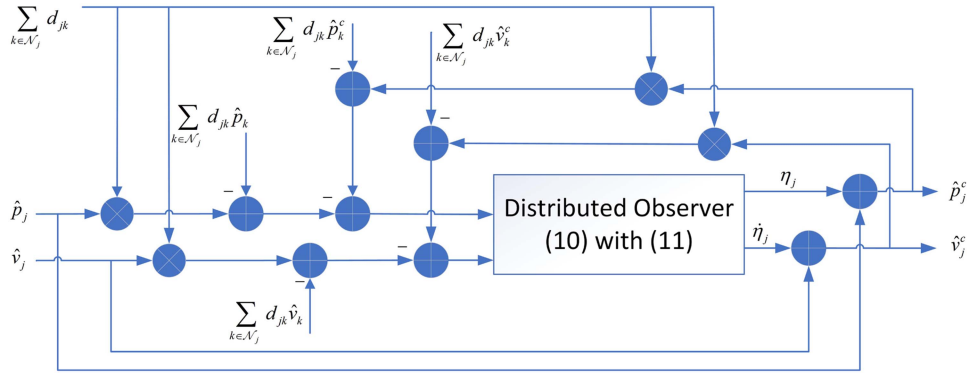


Fig. 2. Signal flow diagram of distributed observer (10) with (11) (Inputs: $\hat{p}_j$, $\hat{v}_j$ and $\hat{p}_k$, $\hat{v}_k$, $k \in \mathcal{N}_j$).

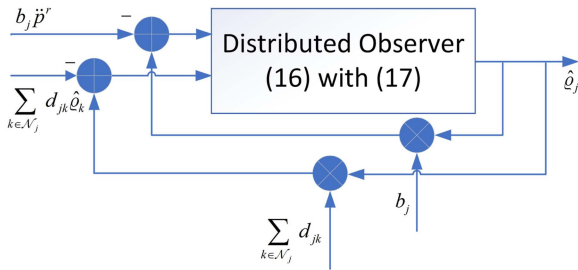


Fig. 3. Signal flow diagram of distributed observer (16) with (17) (Input: $b_j \ddot{p}^r$).

Remark 4: The saturation function (15) can be easily shown to have two useful properties:

- i) $x \text{sat}(x) \geq \|\text{sat}(x)\|^2$; and
- ii) $\text{sat}(x) = 0 \Leftrightarrow x = 0$.

B. Stability Analysis

In the first place, we consider the distributed observer (16) with the adaptive law (17) and demonstrate that the reference acceleration can be estimated accurately for each agent. For this purpose, we first define the estimation error $\tilde{q}_j = \hat{q}_j - \ddot{p}^r$ for each agent $j \in \mathcal{V}$. Using the distributed observer (16), its dynamics can be derived as

to exchange its position p_j , its velocity v_j , and its estimates $\hat{p}_j$, $\hat{v}_j$, $\hat{p}_j^c$, $\hat{v}_j^c$, and $\hat{q}_j$ with its neighbors.

Remark 3: To figure out the configuration centroids of each cluster and the whole team, the initial conditions (9) and (11) are essential. Subject to (9) and (11), all the agents have to coordinate appropriate initial ψ_j , $\dot{\psi}_j$, η_j and $\dot{\eta}_j$. To avoid the resultant over-centralization, these four auxiliary variables are uniformly initialized as 0_d for each agent in practice. Such a setting reinforces the team adaptability to the expansion or contraction of the network scale.

$$\begin{aligned}
 \dot{\tilde{q}}_j &= -\kappa_j \left(\sum_{k \in \mathcal{N}_j} d_{jk} (\hat{q}_j - \hat{q}_k) + b_j (\hat{q}_j - \ddot{p}^r) \right) \\
 &\quad - \hat{\beta}_j \tanh \left(\frac{\sum_{k \in \mathcal{N}_j} d_{jk} (\hat{q}_j - \hat{q}_k) + b_j (\hat{q}_j - \ddot{p}^r)}{\varepsilon(t)} \right) - \ddot{p}^r \\
 &= -\kappa_j \sum_{k=1}^N m_{jk} \tilde{q}_k - \hat{\beta}_j \tanh \left(\frac{\sum_{k=1}^N m_{jk} \tilde{q}_k}{\varepsilon(t)} \right) - \ddot{p}^r,
 \end{aligned} \tag{18}$$

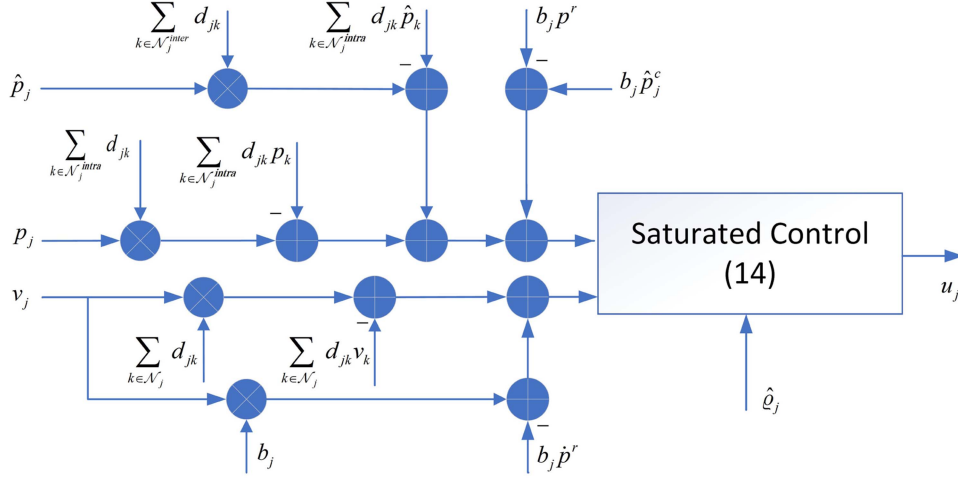


Fig. 4. Signal flow diagram of saturated control (14).

where m_{jk} is the (j, k) -entry of matrix $M = L + \text{diag}\{b_1, b_2, \dots, b_N\}$, and L is the Laplacian matrix associated with the total graph $\mathcal{G}$. Similarly, the adaptive law (17) can be written as

$$\dot{\hat{\beta}}_j = \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \tanh \left(\frac{\sum_{k=1}^N m_{jk} \tilde{q}_k}{\varepsilon(t)} \right). \quad (19)$$

Before giving the convergence result, two useful lemmas are presented first.

Lemma 1 ([36]): Given a connected total graph $\mathcal{G}$, if the reference trajectory is available to at least one agent, then matrix M is positive definite.

Lemma 2 ([37]): For $x \in \mathbb{R}$ and $\varepsilon > 0$, it holds that

$$|x| - x \tanh \left(\frac{x}{\varepsilon} \right) \leq q_e \varepsilon, \quad (20)$$

where q_e satisfies $q_e = e^{-q_e - 1}$, i.e., $q_e = 0.2785$.

The following proposition shows that, using the distributed observer (16) with the adaptive law (17), the estimation error $\tilde{q}_j$ of each agent converges to zero asymptotically, which means that each agent is capable of accurately estimating the reference acceleration.

Proposition 1: Given a connected total graph $\mathcal{G}$, if the reference trajectory is available to at least one agent, the designed distributed observer (16) with the adaptive law (17) guarantees that $\lim_{t \rightarrow \infty} \tilde{q}_j(t) = 0_d$, $j \in \mathcal{V}$.

Proof: Assign a Lyapunov function in the following form:

$$W_e = \tilde{q}^T (M \otimes I_d) \tilde{q} + \frac{1}{2} \sum_{j=1}^N (\hat{\beta}_j - \bar{\beta})^2, \quad (21)$$

where $\tilde{q} = \text{col}(\tilde{q}_1, \tilde{q}_2, \dots, \tilde{q}_N)$ is a column stack of $\tilde{q}_1, \tilde{q}_2, \dots, \tilde{q}_N$, $\otimes$ denotes the Kronecher product, and $\bar{\beta} \geq \|\ddot{p}^r\|_1$ is a constant with $\|\cdot\|_1$ as the 1-norm of a vector. Its derivative along (18) satisfies

$$\dot{W}_e = \sum_{j=1}^N \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \left(-\kappa_j \sum_{k=1}^N m_{jk} \tilde{q}_k \right)$$

$$- \hat{\beta}_j \tanh \left(\frac{\sum_{k=1}^N m_{jk} \tilde{q}_k}{\varepsilon(t)} - \ddot{p}^r \right) + \sum_{j=1}^N (\hat{\beta}_j - \bar{\beta}) \dot{\hat{\beta}}_j. \quad (22)$$

In terms of Lemma 2, we have that

$$\begin{aligned} & \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \left(-\hat{\beta}_j \tanh \left(\frac{\sum_{k=1}^N m_{jk} \tilde{q}_k}{\varepsilon(t)} - \ddot{p}^r \right) \right) \\ & \leq (\bar{\beta} - \hat{\beta}_j) \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \tanh \left(\frac{\sum_{k=1}^N m_{jk} \tilde{q}_k}{\varepsilon(t)} \right) \\ & \quad + \bar{\beta} q_e \varepsilon(t). \end{aligned} \quad (23)$$

Substituting (23) into (22) gives rise to

$$\begin{aligned} \dot{W}_e & \leq \sum_{j=1}^N \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \left(-\kappa_j \sum_{k=1}^N m_{jk} \tilde{q}_k \right) + \sum_{j=1}^N (\hat{\beta}_j - \bar{\beta}) \\ & \quad \left(\dot{\hat{\beta}}_j - \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \tanh \left(\frac{\sum_{k=1}^N m_{jk} \tilde{q}_k}{\varepsilon(t)} \right) \right) + \bar{\beta} q_e \varepsilon(t). \end{aligned} \quad (24)$$

Using (19), it further follows that

$$\begin{aligned} \dot{W}_e & \leq \sum_{j=1}^N \left(\sum_{k=1}^N m_{jk} \tilde{q}_k \right)^T \left(-\kappa_j \sum_{k=1}^N m_{jk} \tilde{q}_k \right) + \bar{\beta} q_e \varepsilon(t) \\ & \leq -\underline{\kappa} \|(M \otimes I_d) \tilde{q}\|^2 + \bar{\beta} q_e \varepsilon(t) \\ & \leq -\underline{\kappa} \mu(M^T M) \|\tilde{q}\|^2 + \bar{\beta} q_e \varepsilon(t), \end{aligned} \quad (25)$$

where $\underline{\kappa} = \min_{j \in \mathcal{V}} \{\kappa_j\}$. According to Lemma 1, $\mu(M^T M)$ denotes the minimum eigenvalue of $M^T M$ that can be shown positive definite. Using Comparison Lemma [38], it thus follows

that

$$W_{\varrho}(t) + \kappa \mu (M^T M) \int_0^{\infty} \|\tilde{\varrho}(\tau)\|^2 d\tau \leq W_{\varrho}(0) + \frac{\bar{\beta} q_e}{\epsilon}, \quad \forall t \geq 0. \quad (26)$$

Hence, $\tilde{\varrho} \in \mathbb{L}_2^1$ and $W_{\varrho} \in \mathbb{L}_{\infty}$.² This, in terms of (21) and Lemma 1, implies that $\tilde{\varrho}, \hat{\beta}_j \in \mathbb{L}_{\infty}, j \in \mathcal{V}$. It next follows from (18) that $\tilde{\varrho} \in \mathbb{L}_{\infty}$. Based on Barbalat Lemma [38], it can be concluded that $\lim_{t \rightarrow \infty} \tilde{\varrho}(t) = 0_{Nd}$, i.e., $\lim_{t \rightarrow \infty} \tilde{\varrho}_j(t) = 0_d, j \in \mathcal{V}$. $\square$

In terms of Proposition 1, we next consider the introduced composite variables s_j in (12) and r_j in (13), and use a proposition to indicate that, driven by the proposed saturated control (14), they each converge to zero asymptotically.

Proposition 2: Consider the composite variables (12) and (13). Given a connected total graph $\mathcal{G}$, if the reference trajectory is available to at least one agent, the proposed saturated control (14) guarantees that $\lim_{t \rightarrow \infty} s_j(t) = 0_d$ and $\lim_{t \rightarrow \infty} r_j(t) = 0_d, j \in \mathcal{V}$.

Proof: First, by considering the definitions of $\hat{p}_j$ and $\hat{v}_j$, the distributed observer (8) amounts to

$$\ddot{\psi}_j = - \sum_{k \in \mathcal{N}_j^{intra}} d_{jk} (\psi_j - \psi_k + \delta_{jk}) - \sum_{k \in \mathcal{N}_j^{intra}} d_{jk} (\dot{\psi}_j - \dot{\psi}_k). \quad (27)$$

Due to the fact that $\delta_{jk} = -\delta_{kj}$ and using the symmetry of the adjacency matrix D , it then follows that $\sum_{j \in \mathcal{V}_i} \ddot{\psi}_j(t) = 0_d, \forall t \geq 0$. Using the initial condition (9), it further follows that

$$\sum_{j \in \mathcal{V}_i} \psi_j(t) = 0_d, \quad \sum_{j \in \mathcal{V}_i} \dot{\psi}_j(t) = 0_d, \quad \forall t \geq 0, \quad i \in \mathcal{M}. \quad (28)$$

With the definitions of $\hat{p}_j$ and $\hat{v}_j$, these thus imply that

$$\sum_{j \in \mathcal{V}_i} \hat{p}_j(t) = \sum_{j \in \mathcal{V}_i} p_j(t), \quad (29)$$

$$\sum_{j \in \mathcal{V}_i} \hat{v}_j(t) = \sum_{j \in \mathcal{V}_i} v_j(t), \quad \forall t \geq 0, \quad i \in \mathcal{M}. \quad (30)$$

Next, come back to (27). Since each intra-cluster graph $\mathcal{G}_i$ is connected, according to [39], it follows that $\lim_{t \rightarrow \infty} (\psi_j(t) - \psi_k(t)) = -\delta_{jk}$ and $\lim_{t \rightarrow \infty} (\dot{\psi}_j(t) - \dot{\psi}_k(t)) = 0_d, j, k \in \mathcal{V}_i, i \in \mathcal{M}$. In view of (28), we thus have that

$$\lim_{t \rightarrow \infty} \dot{\psi}_j(t) = 0_d, \quad j \in \mathcal{V}_i, \quad i \in \mathcal{M}. \quad (31)$$

Using the definitions of $\hat{p}_j$ and $\hat{v}_j$ again, it therefore follows that

$$\lim_{t \rightarrow \infty} (\hat{p}_j(t) - \hat{p}_k(t)) = \lim_{t \rightarrow \infty} (p_j(t) - p_k(t) - \delta_{jk}), \quad j, k \in \mathcal{V}_i, \quad (32)$$

$$\lim_{t \rightarrow \infty} (\hat{v}_j(t) - v_j(t)) = 0_d, \quad j \in \mathcal{V}_i, \quad i \in \mathcal{M}. \quad (33)$$

Likewise, based on the initial condition (11) and using the above arguments, it can be derived that the distributed observer (10)

¹ $\mathbb{L}_2 = \{f : \mathbb{R}^+ \rightarrow \mathbb{R}^n | f \text{ is locally integrable, } \int_0^{\infty} \|f(t)\| dt < \infty\}$

² $\mathbb{L}_{\infty} = \{f : \mathbb{R}^+ \rightarrow \mathbb{R}^n | f \text{ is locally integrable, } \text{ess sup}_{t \in \mathbb{R}^+} \|f(t)\| < \infty\}$

also guarantees

$$\sum_{j=1}^N \hat{p}_j^c = \sum_{j=1}^N \hat{p}_j, \quad (34)$$

$$\lim_{t \rightarrow \infty} \dot{\eta}_j(t) = 0_d, \quad j \in \mathcal{V}, \quad (35)$$

$$\lim_{t \rightarrow \infty} (\hat{p}_j^c(t) - \hat{p}_k^c(t)) = \lim_{t \rightarrow \infty} (\hat{p}_j(t) - \hat{p}_k(t) - \bar{\theta}_{jk}), \quad j, k \in \mathcal{V}, \quad (36)$$

$$\lim_{t \rightarrow \infty} (\hat{v}_j^c(t) - \hat{v}_j(t)) = 0_d, \quad j \in \mathcal{V}. \quad (37)$$

Second, for each agent $j \in \mathcal{V}$, we define the position tracking error $p_j^e = p_j - p^r - \eta_j^r$ and velocity tracking error $v_j^e = v_j - \dot{p}^r$, where $\eta_j^r \in \mathbb{R}^d$ is the expected displacement with respect to the configuration centroid of the whole team that is introduced just for analytical convenience. In terms of (1), their dynamics satisfy

$$\dot{p}_j^e = v_j^e, \quad (38a)$$

$$\dot{v}_j^e = u_j - \ddot{p}^r. \quad (38b)$$

By substituting the proposed saturated control (14) into (38), the following closed-loop system is available:

$$\dot{p}_j^e = v_j^e, \quad (39a)$$

$$\dot{v}_j^e = -\text{sat}_j(c(s_j + r_j)) - \text{sat}_j(cr_j) + \tilde{\varrho}_j. \quad (39b)$$

Let $p^e = \text{col}(p_1^e, p_2^e, \dots, p_N^e)$, $v^e = \text{col}(v_1^e, v_2^e, \dots, v_N^e)$, $s = \text{col}(s_1, s_2, \dots, s_N)$, and $r = \text{col}(r_1, r_2, \dots, r_N)$. Then (39) can be written in the following stack form:

$$\dot{p}^e = v^e, \quad (40a)$$

$$\dot{v}^e = -\text{sat}(c(s + r)) - \text{sat}(cr) + \tilde{\varrho}, \quad (40b)$$

where $\text{sat}(\cdot) = \text{col}(\text{sat}_1(\cdot), \text{sat}_2(\cdot), \dots, \text{sat}_N(\cdot))$. Moreover, we consider the composite variables (12) and (13). Their dynamics satisfy

$$\begin{aligned} \dot{s}_j &= \sum_{k \in \mathcal{N}_j^{intra}} d_{jk} (v_j - v_k) + \sum_{k \in \mathcal{N}_j^{inter}} d_{jk} (v_j - v_k + \dot{\psi}_j - \dot{\psi}_k) \\ &\quad + b_j (v_j + \dot{\psi}_j + \dot{\eta}_j - \dot{p}^r) \\ &= \sum_{k \in \mathcal{N}_j} d_{jk} (v_j - v_k) + b_j (v_j - \dot{p}^r) \\ &\quad + \sum_{k \in \mathcal{N}_j^{inter}} d_{jk} (\dot{\psi}_j - \dot{\psi}_k) + b_j (\dot{\psi}_j + \dot{\eta}_j) \\ &= r_j + \vartheta_j, \end{aligned} \quad (41a)$$

$$\begin{aligned} \dot{r}_j &= \sum_{k \in \mathcal{N}_j} d_{jk} (\dot{v}_j - \dot{v}_k) + b_j (\dot{v}_j - \ddot{p}^r) \\ &= \sum_{k=1}^N m_{jk} \dot{v}_k^e, \end{aligned} \quad (41b)$$

where $\vartheta_j = \sum_{k \in \mathcal{N}_j^{inter}} d_{jk} (\dot{\psi}_j - \dot{\psi}_k) + b_j (\dot{\psi}_j + \dot{\eta}_j)$. By defining $\vartheta = \text{col}(\vartheta_1, \vartheta_2, \dots, \vartheta_N)$ and due to (40), they can be

written in the following stack form:

$$\dot{s} = r + \vartheta, \quad (42a)$$

$$\begin{aligned} \dot{r} &= (M \otimes I_d) \dot{v}^e \\ &= -(M \otimes I_d)(\text{sat}(c(s+r)) + \text{sat}(cr)) + (M \otimes I_d) \tilde{q}, \end{aligned} \quad (42b)$$

which can be regarded as a nominal system

$$\dot{s} = r, \quad (43a)$$

$$\dot{r} = -(M \otimes I_d)(\text{sat}(c(s+r)) + \text{sat}(cr)), \quad (43b)$$

perturbed by $\text{col}(\vartheta, (M \otimes I_d) \tilde{q})$. According to the input-to-state stability theory [38], $\lim_{t \rightarrow \infty} s(t) = 0_{Nd}$ and $\lim_{t \rightarrow \infty} r(t) = 0_{Nd}$ for (42) can be obtained given that

- i) this result holds for the nominal system (43); and
- ii) ϑ and $(M \otimes I_d) \tilde{q}$ asymptotically converge to zero.

Since the asymptotic convergence of ϑ and $(M \otimes I_d) \tilde{q}$ to zero has been ensured by (31), (35) and Proposition 1, we next focus on showing the asymptotic convergence of s and r of the nominal system (43) to zero.

We assign a Lyapunov function in the following form:

$$\begin{aligned} W_s &= \int_0^{c(s+r)} \text{sat}(\iota)^T d\iota + \int_0^{cr} \text{sat}(\iota)^T d\iota \\ &\quad + c(v^e)^T (M \otimes I_d) v^e. \end{aligned} \quad (44)$$

Its derivative along (43) satisfies

$$\begin{aligned} \dot{W}_s &= \text{sat}(c(s+r))^T (cr) - c(\text{sat}(c(s+r)) + \text{sat}(cr))^T \\ &\quad \cdot (M \otimes I_d)(\text{sat}(c(s+r)) + \text{sat}(cr)) \\ &= -c(\text{sat}(c(s+r)) + \text{sat}(cr))^T (M \otimes I_d) \\ &\quad \cdot (\text{sat}(c(s+r)) + \text{sat}(cr)) - (cr)^T \text{sat}(cr) \\ &\leq -c\mu(M) \|\text{sat}(c(s+r)) + \text{sat}(cr)\|^2 - \|\text{sat}(cr)\|^2. \end{aligned} \quad (45)$$

In terms of LaSalle's invariance principle [38], it follows that $\lim_{t \rightarrow \infty} \text{sat}(c(s(t) + r(t))) = 0_{Nd}$ and $\lim_{t \rightarrow \infty} \text{sat}(cr(t)) = 0_{Nd}$. Using the monotonicity of the saturation function $\text{sat}(\cdot)$, these imply that $\lim_{t \rightarrow \infty} (s(t) + r(t)) = 0_{Nd}$ and $\lim_{t \rightarrow \infty} r(t) = 0_{Nd}$, and further that $\lim_{t \rightarrow \infty} s(t) = 0_{Nd}$.

In terms of the above arguments, for the perturbed system (42), we obtain that $\lim_{t \rightarrow \infty} s(t) = 0_{Nd}$ and $\lim_{t \rightarrow \infty} r(t) = 0_{Nd}$, i.e., $\lim_{t \rightarrow \infty} s_j(t) = 0_d$ and $\lim_{t \rightarrow \infty} r_j(t) = 0_d$, $j \in \mathcal{V}$. $\square$

Using Propositions 1 and 2, the main result of this section is given in the following theorem.

Theorem 1: Consider the multi-agent system (1). Given a connected total graph $\mathcal{G}$, if the reference trajectory is available to at least one agent, the proposed saturated control (14) guarantees that the concerned centroid tracking objective (4)–(7) of cluster formation is achieved.

Proof: Due to the fact that $\lim_{t \rightarrow \infty} s_j(t) = 0_d$, $j \in \mathcal{V}$ given by Proposition 2, it follows from (32) and (36) that

$$\lim_{t \rightarrow \infty} \sum_{k \in \mathcal{N}_j^{intra}} d_{jk}(p_j(t) - p_k(t) - \delta_{jk})$$

$$\begin{aligned} &+ \lim_{t \rightarrow \infty} \sum_{k \in \mathcal{N}_j^{inter}} d_{jk}(\hat{p}_j(t) - \hat{p}_k(t) - \theta_{jk}) \\ &+ \lim_{t \rightarrow \infty} b_j(\hat{p}_j^c(t) - p^r(t)) \\ &= \lim_{t \rightarrow \infty} \sum_{k \in \mathcal{N}_j^{intra}} d_{jk}(p_j(t) - p_k(t) - \delta_{jk}) \\ &+ \lim_{t \rightarrow \infty} \sum_{k \in \mathcal{N}_j^{inter}} d_{jk}(\hat{p}_j(t) - \hat{p}_k(t) - \theta_{jk}) \\ &+ \lim_{t \rightarrow \infty} b_j(\hat{p}_j^c(t) - p^r(t)) \\ &= \lim_{t \rightarrow \infty} \sum_{k \in \mathcal{N}_j} d_{jk}(\hat{p}_j(t) - \hat{p}_k(t) - \bar{\theta}_{jk}) \\ &+ \lim_{t \rightarrow \infty} b_j(\hat{p}_j^c(t) - p^r(t)) \\ &= \lim_{t \rightarrow \infty} \left(\sum_{k \in \mathcal{N}_j} d_{jk}(\hat{p}_j^c(t) - \hat{p}_k^c(t)) + b_j(\hat{p}_j^c(t) - p^r(t)) \right) \\ &= 0, \quad j \in \mathcal{V}. \end{aligned} \quad (46)$$

Equivalently, $\lim_{t \rightarrow \infty} (M \otimes I_d)(\hat{p}^c(t) - 1_N \otimes p^r(t)) = 0_{Nd}$, where $\hat{p}^c = \text{col}(\hat{p}_1^c, \hat{p}_2^c, \dots, \hat{p}_N^c)$. According to Lemma 1, this implies that

$$\lim_{t \rightarrow \infty} (\hat{p}_j^c(t) - p^r(t)) = 0_d, \quad j \in \mathcal{V}. \quad (47)$$

In terms of (3), (29), and (34), this further implies that

$$\lim_{t \rightarrow \infty} (p^c(t) - p^r(t)) = 0_d. \quad (48)$$

It next follows from (36) and (47) that $\lim_{t \rightarrow \infty} (\hat{p}_j(t) - \hat{p}_k(t)) = \bar{\theta}_{jk}$, $j, k \in \mathcal{V}$, i.e.,

$$\begin{cases} \lim_{t \rightarrow \infty} (\hat{p}_j(t) - \hat{p}_k(t)) = \theta_{jk}, & j, k \text{ are in the same cluster,} \\ \lim_{t \rightarrow \infty} (\hat{p}_j(t) - \hat{p}_k(t)) = 0_d, & \text{otherwise.} \end{cases} \quad (49)$$

Besides, it follows from (2) and (29) that in each cluster $i \in \mathcal{M}$,

$$\lim_{t \rightarrow \infty} (\hat{p}_j(t) - p_i^c(t)) = 0_d, \quad j \in \mathcal{V}_i. \quad (50)$$

This, together with (49), thus implies that

$$\lim_{t \rightarrow \infty} (p_i^c(t) - p_j^c(t)) = \theta_{ij}, \quad i, j \in \mathcal{M}. \quad (51)$$

In addition, it follows from (32) and (49) that in each cluster $i \in \mathcal{M}$,

$$\lim_{t \rightarrow \infty} (p_j(t) - p_k(t)) = \delta_{jk}, \quad j, k \in \mathcal{V}_i. \quad (52)$$

Furthermore, by considering the fact that $\lim_{t \rightarrow \infty} r_j(t) = 0_d$, $j \in \mathcal{V}$ given by Proposition 2, it follows from (13) that $\lim_{t \rightarrow \infty} (M \otimes I_d)(v(t) - 1_N \otimes \dot{p}^r(t)) = 0_{Nd}$, where $v = \text{col}(v_1, v_2, \dots, v_N)$. This thus implies that

$$\lim_{t \rightarrow \infty} (v_j(t) - \dot{p}_r(t)) = 0_d, \quad j \in \mathcal{V}. \quad (53)$$

Therefore, the concerned centroid tracking objective (4)–(7) of cluster formation is achieved. $\square$

IV. APPLICATION TO CLUSTERED VTOL AAV FORMATION

In this section, we extend the proposed distributed control algorithm in the previous section to the clustered VTOL AAV formation. Suppose that there exists a team of $N \geq 2$ VTOL AAVs, labeled by $\mathcal{V} = \{1, 2, \dots, N\}$, which is separated into n disjoint clusters. In each cluster, one VTOL AAV is authorized as the correspondent. Each VTOL AAV is modeled by

$$\dot{p}_j = v_j, \quad (54)$$

$$\dot{v}_j = \frac{T_j}{m_j} R_j(Q_j) \hat{e}_3 - g \hat{e}_3, \quad (55)$$

$$\dot{Q}_j = \frac{1}{2} G_j(Q_j) \omega_j, \quad (56)$$

$$J_j \dot{\omega}_j = -\omega_j^\times J_j \omega_j + \tau_j, \quad (57)$$

where $p_j = [p_{j,x}, p_{j,y}, p_{j,z}]^T$ and $v_j = [v_{j,x}, v_{j,y}, v_{j,z}]^T$ denote the position and velocity of its fuselage within the inertia frame, m_j is its total mass, $\hat{e}_3 \triangleq [0, 0, 1]^T$, g is the gravitational acceleration, T_j denotes its applied thrust, $Q_j = [\sigma_j, q_j^T]^T \in \mathbb{Q} \triangleq \{Q \in \mathbb{R} \times \mathbb{R}^3 | \sigma^2 + q^T q = 1\}$ denotes its unit quaternion, $R_j(Q_j) = (\sigma_j^2 - q_j^T q_j) I_3 + 2q_j q_j^T - 2\sigma_j q_j^\times$ ³ denotes its rotation matrix, $G_j(Q_j) = [-q_j, \sigma_j I_3 - q_j^\times]^T$, $\omega_j = [\omega_{j,x}, \omega_{j,y}, \omega_{j,z}]^T$ denotes its angular velocity in its body frame, $J_j = \text{diag}\{J_{j,x}, J_{j,y}, J_{j,z}\}$ denotes its inertial matrix with respect to its body frame, and $\tau_j = [\tau_{j,x}, \tau_{j,y}, \tau_{j,z}]^T$ denotes its applied torque within its body frame.

Given a reference trajectory $p^r = [p_x^r, p_y^r, p_z^r]^T$, we focus on the fulfillment of the centroid tracking objective (4)–(7) of clustered VTOL AAV formation. For this purpose, as done in [11], we introduce a hierarchical structure for the distributed control algorithm design, under which the outer position loop and inner attitude loop can be exploited in sequence.

In the first place, in the outer position loop, using the distributed observers (8), (10), and (16) with initial conditions (9) and (11), we design an intermediate command force $u_j^c = T_j R_j(Q_j^c) \hat{e}_3 / m_j$ for each VTOL AAV in the following form:

$$u_j^c = -\text{sat}_j(c(s_j + r_j)) - \text{sat}_j(cr_j) + \hat{p}_j + g \hat{e}_3, \quad (58)$$

where the parameter definitions are the same with those in (14). Thus, the applied thrust T_j of each VTOL AAV can be obtained as

$$T_j = m_j \|u_j^c\|. \quad (59)$$

Next, motivated by [21], we extract the command attitude $Q_j^c = [\sigma_j^c, (q_j^c)^T]^T$ from the command force u_j^c as follows:

$$\sigma_j^c = \sqrt{\frac{1}{2} + \frac{g - u_{j,z}^c}{2\|u_j^c\|}}, \quad q_j^c = \frac{1}{2\|u_j^c\| \sigma_j^c} \begin{bmatrix} u_{j,y}^c \\ -u_{j,x}^c \\ 0 \end{bmatrix}. \quad (60)$$

³Superscript “ $\times$ ” defines a transformation from $x = [x_1, x_2, x_3]^T$ into a skew-symmetric matrix $x^\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$.

Also, according to [21], the non-singularity of such an extraction can be guaranteed given that the saturation bounds $\bar{\alpha}_j$ and $\underline{\alpha}_j$ satisfy $g - 2 \max\{\bar{\alpha}_j, -\underline{\alpha}_j\} - \lambda_j > 0$.

In the second place, in the inner attitude loop, we define the attitude error $Q_j^e = [\sigma_j^e, (q_j^e)^T]^T = (Q_j^c)^{-1} \odot Q_j$ ⁴ and angular velocity error $\omega_j^e = \omega_j - R_j(Q_j^e)^T \omega_j^c$. To achieve the attitude tracking, we implement the applied torque τ_j proposed in [21] for each VTOL AAV as follows:

$$\begin{aligned} \tau_j = & -c_q q_j^e - c_\omega \omega_j^e + (R_j(Q_j^e)^T \omega_j^e)^\times J_j R_j(Q_j^e)^T \omega_j^c \\ & + J_j R_j(Q_j^e)^T \dot{\omega}_j^c, \end{aligned} \quad (61)$$

where $c_q > 0$, $c_\omega > 0$ are two control parameters, and the expressions of the command angular velocity ω_j^c and its derivative $\dot{\omega}_j^c$ are given in [40].

The following theorem shows that the built command force and applied torque guarantee the achievement of the centroid tracking objective of clustered VTOL AAV formation.

Theorem 2: Consider a team of VTOL AAVs (54)–(57). Given a connected total graph $\mathcal{G}$, if the reference trajectory is available to at least one VTOL AAV, the proposed command force (58) and applied torque (61) guarantee that the centroid tracking objective (4)–(7) of clustered VTOL AAV formation is achieved.

Proof: According to [21], it can be shown that the applied torque (61) guarantees the accurate attitude tracking of each quad-rotor, i.e., $\lim_{t \rightarrow \infty} q_j^e(t) = 0_3$ and $\lim_{t \rightarrow \infty} \omega_j^e(t) = 0_3, j \in \mathcal{V}$. Next, recall the definitions of the composite variables s_j and r_j in (12) and (13). Using the similar derivations as given in (41), it can be deduced that their stacks satisfy the following dynamics:

$$\dot{s} = r + \vartheta, \quad (62a)$$

$$\dot{r} = -(M \otimes I_d)(\text{sat}(c(s+r)) + \text{sat}(cr)) + (M \otimes I_d)(\tilde{\rho} + \xi), \quad (62b)$$

where the parameter definitions are the same with those in (42) except that $\xi = \text{col}(\xi_1, \xi_2, \dots, \xi_N)$ with $\xi_j = T_j(R_j(Q_j) - R_j(Q_j^c)) \hat{e}_3 / m_j$. (62) can be treated as the nominal system (43) perturbed by $\text{col}(\vartheta, (M \otimes I_d)(\tilde{\rho} + \xi))$. According to the input-to-state stability theory [38], $\lim_{t \rightarrow \infty} s(t) = 0_{3N}$ and $\lim_{t \rightarrow \infty} r(t) = 0_{3N}$ are available given that

- i) this result holds for the nominal system (43); and
- ii) ϑ and $(M \otimes I_d)(\tilde{\rho} + \xi)$ asymptotically converge to zero.

First, by taking the Lyapunov function (44) and following the arguments after (44) in Proposition 2, $\lim_{t \rightarrow \infty} s(t) = 0_{3N}$ and $\lim_{t \rightarrow \infty} r(t) = 0_{3N}$ for the nominal system (43) can be shown trivially. Next, by following the arguments in Proposition 1 and those from (27) to (37) in Proposition 2, the asymptotic convergence of ϑ and $(M \otimes I_d)\tilde{\rho}$ to zero can also be obtained. Next, it can be shown that $(R_j(Q_j) - R_j(Q_j^c)) \hat{e}_3 = R_j(Q_j^c)(R_j(Q_j^e) - I_3) \hat{e}_3 = \nu_j^\times q_j^e$ ⁵ with $\nu_j = [-q_{j,y}^e, q_{j,x}^e, \sigma_j^e]^T$. Besides, in terms of (59), the saturation of

⁴Operator “ $\odot$ ” denotes the product of unit quaternion which is defined by $Q_1 \odot Q_2 = [\sigma_1 \sigma_2 - q_1^T q_2, (\sigma_1 q_2 + \sigma_2 q_1 + S(q_1) q_2)^T]^T$ for $Q_1 = [\sigma_1, q_1^T]^T$ and $Q_2 = [\sigma_2, q_2^T]^T$.

⁵ $R_j(Q_j) = R_j(Q_j^c) R_j(Q_j^e)$ [41]

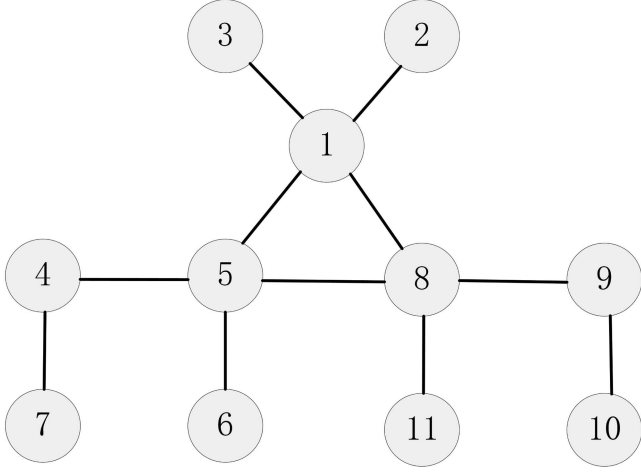


Fig. 5. Communication topology of VTOL AAVs (Example 1).

the designed command force (58) ensures the boundedness of the applied thrust T_j . Accordingly, each ξ_j asymptotically converges to zero, and so does $(M \otimes I_d)\xi$. Consequently, for the perturbed system (62), we obtain that $\lim_{t \rightarrow \infty} s(t) = 0_{3N}$ and $\lim_{t \rightarrow \infty} r(t) = 0_{3N}$, i.e., $\lim_{t \rightarrow \infty} s_j(t) = 0_3$ and $\lim_{t \rightarrow \infty} r(t) = 0_3, j \in \mathcal{V}$.

Furthermore, by following the arguments in Theorem 1, the achievement of the centroid tracking objective (4)–(7) of clustered VTOL AAV formation can be shown. $\square$

V. SIMULATION VERIFICATION

In this section, we first verify effectiveness of the proposed distributed control algorithm on a team of VTOL AAVs. Then, we substantiate its adaptation capability to team scale expansion.

A. Example 1: Effectiveness Verification

There are three clusters of VTOL AAVs, which, respectively, include three, four and four members. Their communication topology is depicted in Fig. 5, which can be confirmed to be connected. Also, note from Fig. 5 that AAVs 1, 5, and 8 serve as correspondents. The inertial parameters and saturation bounds of the AAVs are given in Table I. The reference trajectory is formulated as $p^r(t) = [10 \cos(0.1t) + 0.9t, 10 \sin(0.1t) + 0.9t, 0.2t]^T$ (m), which is just known to the first VTOL AAV. The expected configuration of cluster 1 is given by $\delta_{12} = [3, -3\sqrt{3}, 0]^T$ (m) and $\delta_{31} = [3, 3\sqrt{3}, 0]^T$ (m), and the expected configuration of cluster 2 (or 3) is given by δ_{45} (or δ_{89}) $= [-6, 0, 0]^T$ (m), δ_{56} (or δ_{910}) $= [0, -6, 0]^T$ (m), and δ_{74} (or δ_{118}) $= [0, 6, 0]^T$ (m). Besides, the expected inter-cluster configuration is given by $\theta_{12} = [0, -18, 0]^T$ (m), $\theta_{23} = [-18, 18, 0]^T$ (m), and $\theta_{31} = [18, 0, 0]^T$ (m). Moreover, the initial positions are, respectively, $p_1 = [4, 4, 1]^T$ m, $p_2 = [3, -4, 2]^T$ m, $p_3 = [5, 3, -1]^T$ m, $p_4 = [2, -2, -4]^T$ m, $p_5 = [-2, 1, 2]^T$ m, $p_6 = [-2, -2, 2]^T$ m, $p_7 = [-5, 2, -1]^T$ m, $p_8 = [-1, -1, 0]^T$ m, $p_9 = [2, 2, 2]^T$ m, $p_{10} = [-1, -1, 2]^T$ m, and $p_{11} = [3, 0, 1]^T$ m. And other system states as well auxiliary and adaptive variables are initialized as zero.

TABLE I
INITIAL SYSTEM STATES AS WELL AS AUXILIARY AND ADAPTIVE VARIABLES

UAVs	m_j (kg)	J_j (kgm ²)	$\bar{\alpha}_j$	$\underline{\alpha}_j$
1	0.9	$\text{diag}\{4.8, 4.8, 8.8\} * 10^{-2}$	4	-2
2	1.3	$\text{diag}\{0.15, 0.15, 0.3\}$	3	-2
3	0.3	$\text{diag}\{2.5, 2.5, 5\} * 10^{-2}$	2	-3
4	0.6	$\text{diag}\{4.6, 4.6, 8.2\} * 10^{-2}$	4	-2
5	1	$\text{diag}\{0.13, 0.13, 0.2\}$	3	-2
6	0.5	$\text{diag}\{3, 3, 6\} * 10^{-2}$	2	-4
7	3	$\text{diag}\{0.3, 0.3, 0.6\}$	2	-4
8	0.3	$\text{diag}\{0.15, 0.15, 0.3\}$	2	-3
9	3.5	$\text{diag}\{2.5, 2.5, 5\} * 10^{-2}$	2	-3
10	4	$\text{diag}\{0.36, 0.36, 0.66\}$	3	-2
11	0.6	$\text{diag}\{4.6, 4.6, 8.5\} * 10^{-2}$	4	-3

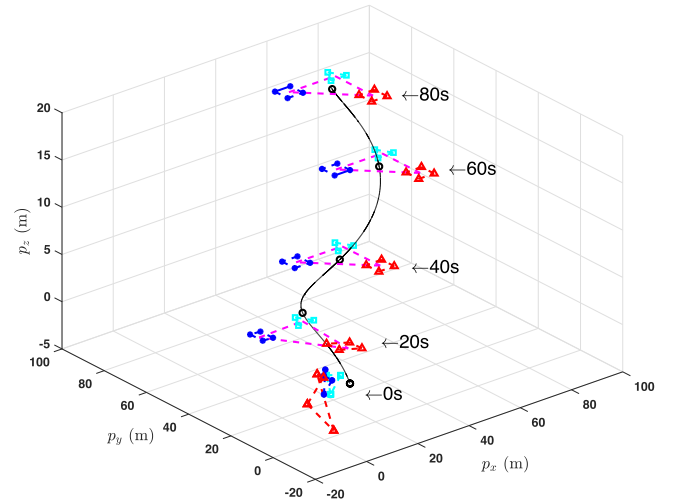


Fig. 6. Snapshots of formation evolution (Cyan dashed line: Configuration of cluster 1, Blue dashed line: Configuration of cluster 2, Red dashed line: Configuration of cluster 3, Pink dashed line: Configuration of centroids, Black solid line: reference trajectory, Black circle: Centroid of whole team).

In addition, the control and adaptive parameters are chosen as $c = 4$ and $\kappa_j = \lambda_j = 0.5, j \in \{1, 2, \dots, 11\}$. The simulation results are illustrated in Figs. 6–9.

Fig. 6 depicts the three-dimensional formation evolution of the whole VTOL AAV team, in which the formation is given at regular 20 s intervals. It can be seen therein that after nearly 40 s, three clusters, respectively, form the prespecified configurations, their centroids also from the prespecified configuration, and the centroid of the whole team tracks the reference trajectory accurately. This validates the achievement of the centroid tracking objective of clustered VTOL AAV formation. Specifically, Fig. 7 illustrates the trajectories of position tracking errors with respect to the reference trajectory. It can be observed therein that they all converge to zero asymptotically. This indicates that each VTOL AAV accurately reaches the position within its corresponding configuration. Besides, Fig. 8 shows the tracking errors of the configuration centroid of the whole team with respect to the reference trajectory. It can be observed therein that they all converge to zero asymptotically,

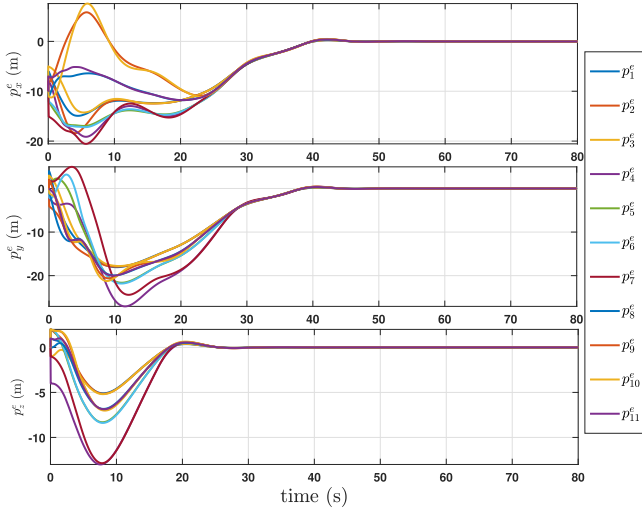


Fig. 7. Trajectories of position tracking errors (Example 1).

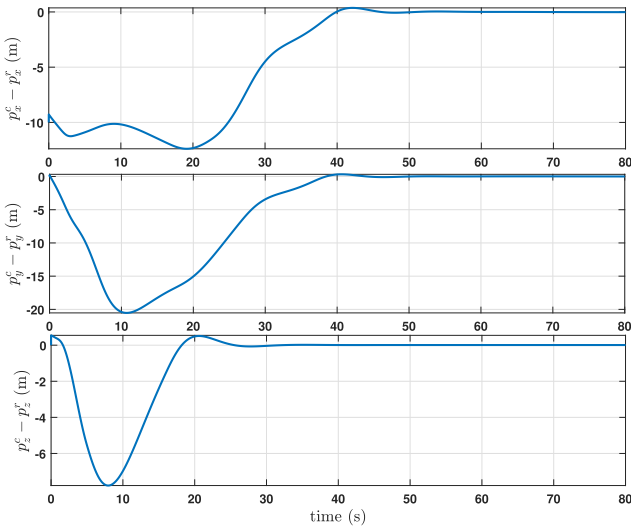


Fig. 8. Trajectories of tracking errors of configuration centroid (Example 1).

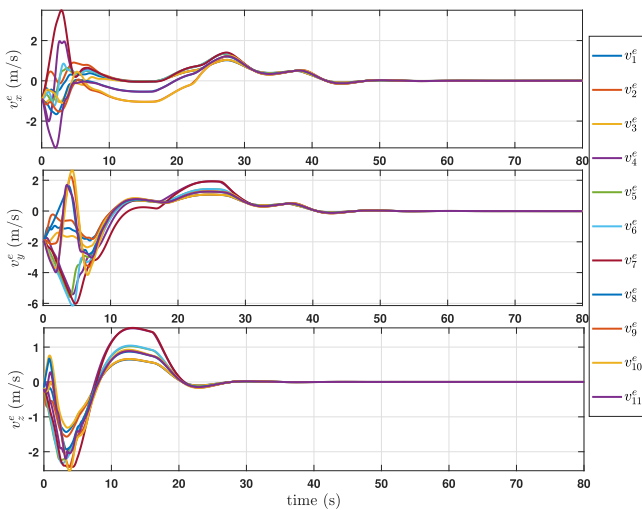


Fig. 9. Trajectories of velocity tracking errors (Example 1).

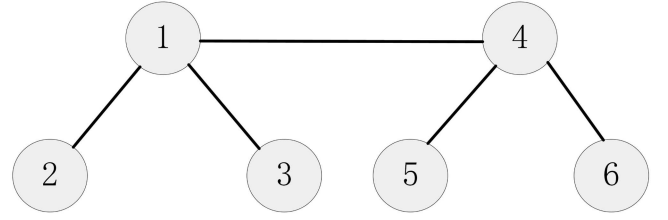


Fig. 10. Communication topology of VTOL AAVs within the first two clusters (Example 2).

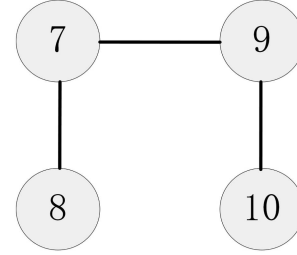


Fig. 11. Communication topology of VTOL AAVs within the third cluster (Example 2).

which demonstrates the accurate centroid tracking even though each AAV has no ideas of the centroid information. Finally, Fig. 9 exhibits that the velocity tracking errors converge to zero asymptotically.

B. Example 2: Scalability Verification

In the initial stage, there are two clusters of VTOL AAVs, which both include three members. Their communication topology is depicted in Fig. 10, which is connected. Their inertial parameters are the same, i.e., $m_j = 0.9$ kg and $J_j = \text{diag}\{4.8, 4.8, 8.8\} * 10^{-2}$ kgm², $j = 1, 2, \dots, 6$. The reference trajectory herein is formulated as $p^r(t) = [10 \cos(0.1t), 0.3t, 4]^T$ (m), which is known to AAVs 1 and 4. The expected configurations of these two clusters are given by δ_{12} (or δ_{45}) = $[-1, 1, 0]^T$ (m) and δ_{13} (or δ_{46}) = $[1, 1, 0]^T$ (m). Besides, the expected configuration between two clusters is given by $\theta_{12} = [-3, 3, 0]^T$ (m). Moreover, the initial positions are, respectively, $p_j = [1, 1, 0]^T * j$ (m), $j = 1, 2, \dots, 6$. The other initial states and parameters are the same with those in the above example. To verify the scalability competence of the team using the saturated distributed control algorithm, we suppose that another cluster of four VTOL AAVs joins the team at 120 s. The communication topology within this cluster is depicted in Fig. 11, and a new communication between AAVs 1 and 7 is built such that the overall topology is connected. Besides, the expected configuration of this new cluster is given by $\delta_{78} = [0, 2, 0]^T$ (m), $\delta_{79} = [2, 2, 0]^T$ (m), and $\delta_{910} = [0, -2, 0]^T$ (m). Moreover, the expected configuration between this new cluster and the original clusters is given by $\theta_{13} = [3, -3, 0]^T$ (m) and $\theta_{23} = [6, 0, 0]^T$ (m). Fig. 12 shows the planar snapshot of the formation centroid tracking. It can be seen therein that, before 120 s, two original clusters form their expected configurations, respectively, the centroids of these two clusters also form their expected configuration, and the centroid

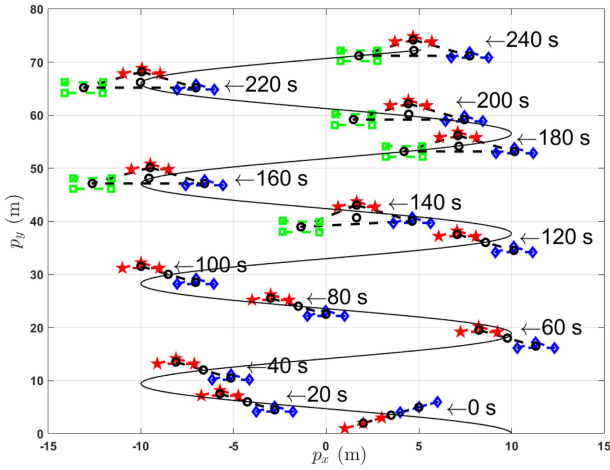


Fig. 12. Planar snapshot of formation centroid tracking (Red dashed line: Configuration of cluster 1, Blue dashed line: Configuration of cluster 2, Green dashed line: Configuration of cluster 3, Black dashed line: Configuration of centroids, Black solid line: reference trajectory, Black circle: Centroids of clusters and team).

of the whole team tracks the reference trajectory. When the new cluster joins the team at 120 s, it can be seen in Fig. 12 that the new cluster forms its expected configuration without affecting the configurations of the original two clusters, the centroids of three clusters form the new expected configuration, and the new centroid of the whole team tracks the reference trajectory. This substantiates that the proposed saturated distributed control algorithm is of favorable adaptation to the online team scale expansion.

VI. CONCLUSION

This paper proposes a saturated distributed control algorithm for the centroid tracking of cluster formation. Particularly, we first design two distributed observers such that, under proper initial conditions, each agent is able to obtain the centroid of its corresponding cluster and that of the whole team accurately. Next, we design another distributed observer to enable all the agents to obtain the reference acceleration information accurately. Using the outputs from the distributed observers, a saturated input is designed for each agent such that the multi-agent system achieves the centroid tracking objective of cluster formation. The proposed saturated distributed control algorithm allows heterogeneous saturation levels. The detailed stability analysis is also undertaken under the Lyapunov framework. Furthermore, the proposed saturated distributed control algorithm is successfully applied to the centroid tracking of clustered VTOL AAV formation. In the future, we will focus on studying specific separation algorithms for clustering agents and collision-avoidance-based distributed control algorithms for the centroid tracking control of cluster formation. Also, we will analyze the effect caused by sensor noise in the ultimate convergence accuracy.

REFERENCES

- [1] B. D. O. Anderson, C. Yu, B. Fidan, and J. M. Hendrickx, "Rigid graph control architectures for autonomous formations," *IEEE Control Syst. Mag.*, vol. 28, no. 6, pp. 48–63, Dec. 2008.
- [2] Z. Sun, *Cooperative Coordination and Formation Control for Multi-Agent Systems*. Cham, Switzerland: Springer, 2018.
- [3] E. M. Cruz, L. R. G. Carrillo, and E. S. E. Quesada, "Triangular formation cluster space control for multi-agent entrapment/escorting tasks," in *2023 IEEE Conf. Control Technol. Appl.*, Bridgetown, Barbados, 2023, pp. 59–64.
- [4] G. A. G. Reyes, L. R. G. Carrillo, and P. Rangel, "A geometric control strategy for real-time coordination of multiple unmanned aircraft systems," in *2018 Int. Conf. Unmanned Aircr. Syst.*, Dallas, TX, USA, 2018, pp. 1379–1387.
- [5] K. Oh, M. Park, and H. Ahn, "A survey of multi-agent formation control," *Automatica*, vol. 53, pp. 424–440, 2015.
- [6] M. Trinh, S. Zhao, Z. Sun, D. Zelazo, B. D. O. Anderson, and H. Ahn, "Bearing-based formation control of a group of agents with leader-first follower structure," *IEEE Trans. Autom. Control*, vol. 64, no. 2, pp. 598–613, Feb. 2019.
- [7] Y. Lin, Z. Lin, Z. Sun, and B. D. O. Anderson, "A unified approach for finite-time global stabilization of affine, rigid, and translational formation," *IEEE Trans. Autom. Control*, vol. 67, no. 4, pp. 1869–1881, Apr. 2022.
- [8] T. Ding, M. Ge, C. Xiong, Z. Liu, and G. Ling, "Prescribed-time formation tracking of second-order multi-agent networks with directed graphs," *Automatica*, vol. 152, 2023, Art. no. 110997.
- [9] Y. Liu and Z. Liu, "Distributed adaptive formation control of multi-agent systems with measurement noises," *Automatica*, vol. 150, 2023, Art. no. 110857.
- [10] F. Chen and D. V. Dimarogonas, "Leader-follower formation control with prescribed performance guarantees," *IEEE Trans. Autom. Control*, vol. 8, no. 1, pp. 450–461, Mar. 2021.
- [11] Y. Zou, H. Zhang, and W. He, "Adaptive coordinated formation control of heterogeneous vertical takeoff and landing UAVs subject to parametric uncertainties," *IEEE Trans. Cybern.*, vol. 52, no. 5, pp. 3184–3195, May 2022.
- [12] Y.-H. Su, P. Bhowmick, and A. Lanzon, "A robust adaptive formation control methodology for networked multi-agent UAV systems with applications to cooperative payload transportation," *Control Eng. Pract.*, vol. 138, 2023, Art. no. 105608.
- [13] Y.-H. Su, P. Bhowmick, and A. Lanzon, "Properties of interconnected negative imaginary systems and extension to formation-containment control of networked multi-UAV systems with experimental validation results," *Asian J. Control*, 2023, doi: 10.1002/asjc.3258.
- [14] Y.-H. Su, P. Bhowmick, and A. Lanzon, "A fixed-time formation-containment control scheme for multi-agent systems with motion planning: Applications to quadcopter UAVs," *IEEE Trans. Veh. Technol.*, vol. 73, no. 7, pp. 9495–9507, Jul. 2024.
- [15] Y. Zou and Z. Meng, "Coordinated trajectory tracking of multiple vertical take-off and landing UAVs," *Automatica*, vol. 99, pp. 33–40, 2019.
- [16] M. A. Maghenem, A. Loria, and E. Panteley, "Cascades-based leader-follower formation tracking and stabilization of multiple nonholonomic vehicles," *IEEE Trans. Autom. Control*, vol. 65, no. 8, pp. 3639–3646, Aug. 2020.
- [17] W. Liu, X. Wang, and S. Li, "Formation control for leader-follower wheeled mobile robots based on embedded control technique," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 1, pp. 265–280, Jan. 2023.
- [18] S. Dai, S. He, H. Lin, and C. Wang, "Platoon formation control with prescribed performance guarantees for USVs," *IEEE Trans. Ind. Electron.*, vol. 65, no. 5, pp. 4237–4246, May 2018.
- [19] B. S. Park and S. J. Yoo, "An error transformation approach for connectivity-preserving and collision-avoiding formation tracking of networked uncertain underactuated surface vessels," *IEEE Trans. Cybern.*, vol. 49, no. 8, pp. 2955–2966, Aug. 2019.
- [20] S. He, M. Wang, S. Dai, and F. Luo, "Leader-follower formation control of USVs with prescribed performance and collision avoidance," *IEEE Trans. Ind. Informat.*, vol. 15, no. 1, pp. 572–581, Jan. 2019.
- [21] Y. Sun, Y. Zou, X. He, Q. Fu, and J. Wu, "Distributed formation centroid tracking control of clustered rotorcraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 2, pp. 1867–1878, Apr. 2023.
- [22] T. Hu and Z. Lin, *Control Systems With Actuator Saturation: Analysis and Design*. Boston, MA, USA: Birkhäuser, 2001.
- [23] T. Yang, Z. Meng, D. V. Dimarogonas, and K. H. Johansson, "Global consensus for discrete-time multi-agent systems with input saturation constraints," *Automatica*, vol. 50, no. 2, pp. 499–506, 2014.
- [24] L. Ding, W. Zheng, and G. Guo, "Network-based practical set consensus of multi-agent systems subject to input saturation," *Automatica*, vol. 89, pp. 316–324, 2018.

- [25] X. You, C. Hua, and X. Guan, "Event-triggered leader-following consensus for nonlinear multiagent systems subject to actuator saturation using dynamic output feedback method," *IEEE Trans. Autom. Control*, vol. 63, no. 12, pp. 4391–4396, Dec. 2018.
- [26] X. Yi, T. Yang, J. Wu, and K. H. Johansson, "Distributed event-triggered control for global consensus of multi-agent systems with input saturation," *Automatica*, vol. 63, no. 12, pp. 1–9, 2019.
- [27] X. Wang, H. Su, X. Wang, and G. Chen, "Fully distributed event-triggered semiglobal consensus of multi-agent systems with input saturation," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 5055–5064, Jun. 2017.
- [28] Z. Wang, Y. Zou, Y. Liu, and Z. Meng, "Distributed control algorithm for leader-follower formation tracking of multiple quadrotors: Theory and experiment," *IEEE/ASME Trans. Mechatron.*, vol. 26, no. 2, pp. 1095–1105, Apr. 2021.
- [29] Y. Wang, G. Yu, W. Xie, W. Zhang, and C. Silvestre, "Robust saturated formation tracking control of multiple quadrotors with switching communication topologies," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 6, pp. 3744–3753, Nov./Dec. 2023.
- [30] G. Chen, D. Yao, H. Li, Q. Zhou, and R. Lu, "Saturated threshold event-triggered formation tracking of multiagent systems under sensor attacks and its application to UAVs," *IEEE Trans. Circuits Syst. I: Reg. Papers*, vol. 69, no. 2, pp. 884–895, Feb. 2022.
- [31] Y. Chen, M. Chiang, and L. Fu, "Three-dimensional maneuver control of multiagent systems with constrained input," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6133–6145, Oct. 2023.
- [32] J. Fu, Y. Lv, G. Wen, and X. Yu, "Local measurement based formation navigation of nonholonomic robots with globally bounded inputs and collision avoidance," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 3, pp. 2342–2354, Jul–Sep. 2021.
- [33] L. Consolini, F. Morbidi, D. Prattichizzo, and M. Tosques, "Leader-follower formation control of nonholonomic mobile robots with input constraints," *Automatica*, vol. 44, no. 5, pp. 1343–1349, 2008.
- [34] Y. Zou, Z. Zuo, K. Xia, and M. V. Basin, "Saturated sampled-data distributed control for interval consensus of multi-agent systems," *IEEE Trans. Signal Inf. Process. Over Netw.*, vol. 8, pp. 1024–1036, 2022.
- [35] Z. Zuo, J. Ji, Z. Zhang, Y. Wang, and W. Zhang, "Consensus of double-integrator multi-agent systems with asymmetric input saturation," *Syst. Control Lett.*, vol. 172, 2023, Art. no. 105440.
- [36] Z. Qu, *Cooperative Control of Dynamical Systems: Applications to Autonomous Vehicles*, Berlin, Germany: Springer, 2009.
- [37] M. Chen, S. S. Ge, and B. Ren, "Adaptive tracking control of uncertain MIMO nonlinear systems with input constraints," *Automatica*, vol. 47, no. 3, pp. 452–465, 2011.
- [38] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 2002.
- [39] W. Ren, R. W. Beard, and E. M. Atkins, "Information consensus in multivehicle cooperative control," *IEEE Control Syst. Mag.*, vol. 27, no. 2, pp. 71–82, Apr. 2007.
- [40] Y. Zou, "Nonlinear robust adaptive hierarchical sliding mode control approach for quadrotors," *Int. J. Robust Nonlinear Control*, vol. 27, no. 6, pp. 925–941, 2017.
- [41] M. D. Shuster, "A survey of attitude representations," *J. Astronautical Sci.*, vol. 41, no. 4, pp. 439–517, 1993.



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